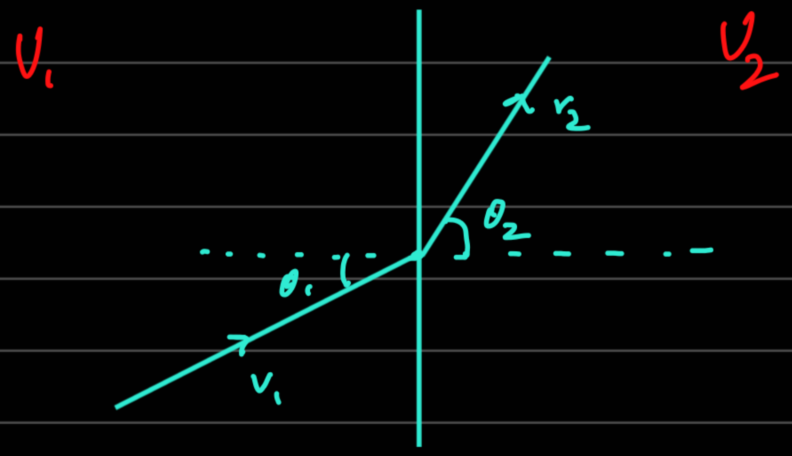
1. *Utilizing conservation of energy and momentum:* A particle of mass m moving with a velocity $\boldsymbol{v}_1$ leaves a half-space in which its potential energy is a constant, say, U_1, and enters another in which its potential energy us a different constant, say, U_2. Using the laws of conservation of energy and momentum, determine the change in the direction of motion of the particle.



$$v_1 \sin\theta_1 = v_2 \sin\theta_2 \quad - \quad 1$$

$$\frac{1}{2} m v_1^2 + U_1 = \frac{1}{2} m v_2^2 + U_2 \quad - \quad 2$$

$$\frac{1}{2} m v_1^2 + U_1 = \frac{1}{2} m v_1^2 \frac{\sin^2\theta_1}{\sin^2\theta_2} + U_2$$

$$\frac{m v_1^2}{2}\left(1 - \frac{\sin^2\theta_1}{\sin^2\theta_2}\right) = U_2 - U_1$$

$$1 - \frac{\sin^2\theta_1}{\sin^2\theta_2} = \frac{2(U_2 - U_1)}{m v_1^2}$$

$$\boxed{\frac{\sin\theta_1}{\sin\theta_2} = \sqrt{1 - \frac{2(U_2 - U_1)}{m v_1^2}}}$$

2. *Magnitude of angular momentum:* Obtain the expressions for the Cartesian components and the magnitude of the angular momentum of a particle in the cylindrical and the spherical polar coordinates.

$$\bar{M} = \bar{r} \times \bar{p}$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & y & z \\ \dot{x} & \dot{y} & \dot{z} \end{vmatrix}$$

Cartesian

$$M_x = m(y\dot{z} - \dot{y}z) \qquad M_y = m(\dot{x}z - x\dot{z})$$

$$M_z = m(x\dot{y} - \dot{x}y)$$

Cylindrical

$$\bar{r} = \rho\hat{\rho} + z\hat{z}$$

$$\bar{v} = \dot{\rho}\hat{\rho} + \rho\dot{\theta}\hat{\theta} + \dot{z}\hat{z}$$

$$m(\bar{r} \times \bar{v}) = m\left(\rho^2\dot{\theta}\hat{z} - \rho\dot{z}\hat{\theta} + \dot{\rho}z\hat{\theta} - \rho z\dot{\theta}\hat{\rho}\right)$$

$$\therefore \boxed{M_\rho = -m\rho z\dot{\theta} \qquad M_\theta = m\dot{\rho}z - m\rho\dot{z} \qquad M_z = m\rho^2\dot{\theta}}$$

Spherical

$$\bar{r} = r\hat{r}$$

$$\bar{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\dot{\phi}\hat{\phi}$$

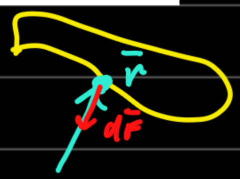
$$m(\bar{r} \times \bar{v}) = m\left(r^2\dot{\theta}\hat{\phi} - r^2\sin\theta\dot{\phi}\hat{\theta}\right)$$

$$\therefore \boxed{M_r = 0 \qquad M_\theta = -mr^2\sin\theta\dot{\phi} \qquad M_\phi = mr^2\dot{\theta}}$$

4. *Perfect gas law:* A perfect gas is defined as one for which the forces of interaction between the molecules of the gas are negligible. This occurs, for example, when the gas is so dilute that the collisions between the molecules are rare when compared to the collisions with the walls of the container. Using these information and the virial theorem, obtain the perfect gas law, viz.

$$P\,V = N\,k_{\mathrm{B}}\,T,$$

where P, V, T and N denote the pressure, the volume, the temperature and the number of molecules of the gas, respectively, and k_{B} is the Boltzmann constant.



$$\langle K.E.\rangle = \frac{3}{2} N k_B T$$

$$\bar{r}\cdot d\bar{F} = -\bar{r}\cdot\hat{n}\, P\, dA$$

$$\Rightarrow \int \bar{r}\cdot d\bar{F} = -\int_S \bar{r}\cdot\hat{n}\, P\, dA$$

$$\left\langle \int \bar{r}\cdot d\bar{F}\right\rangle = -P\int_V (\nabla\cdot\bar{r})\, dV = -3PV$$

$$\therefore \quad 3Nk_BT = 3PV \quad \text{or} \quad PV = Nk_BT$$